Smart Default Detection in Electric Vehicles

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01; Introduction

Task

Global EV(BEV + PHEV) Trend

EV adoption is rapidly increasing worldwide.

'17

EV adoption has grown rapidly over the last decade.

'23

From 2017 to 2023, sales increased from **1.5M to 14.1M** units, with a **CAGR of 45.8%.**

~'2024.02

By early 2024, EV demand rose by **24.4%** compared to the previous year.

14,073 k

+24.4%

(Compared to ~'23.02)

RE 100

: Global EV and Battery Monthly Tracker – Mar 2024, SNE

Emerging Risk
In the EV Era

Motor

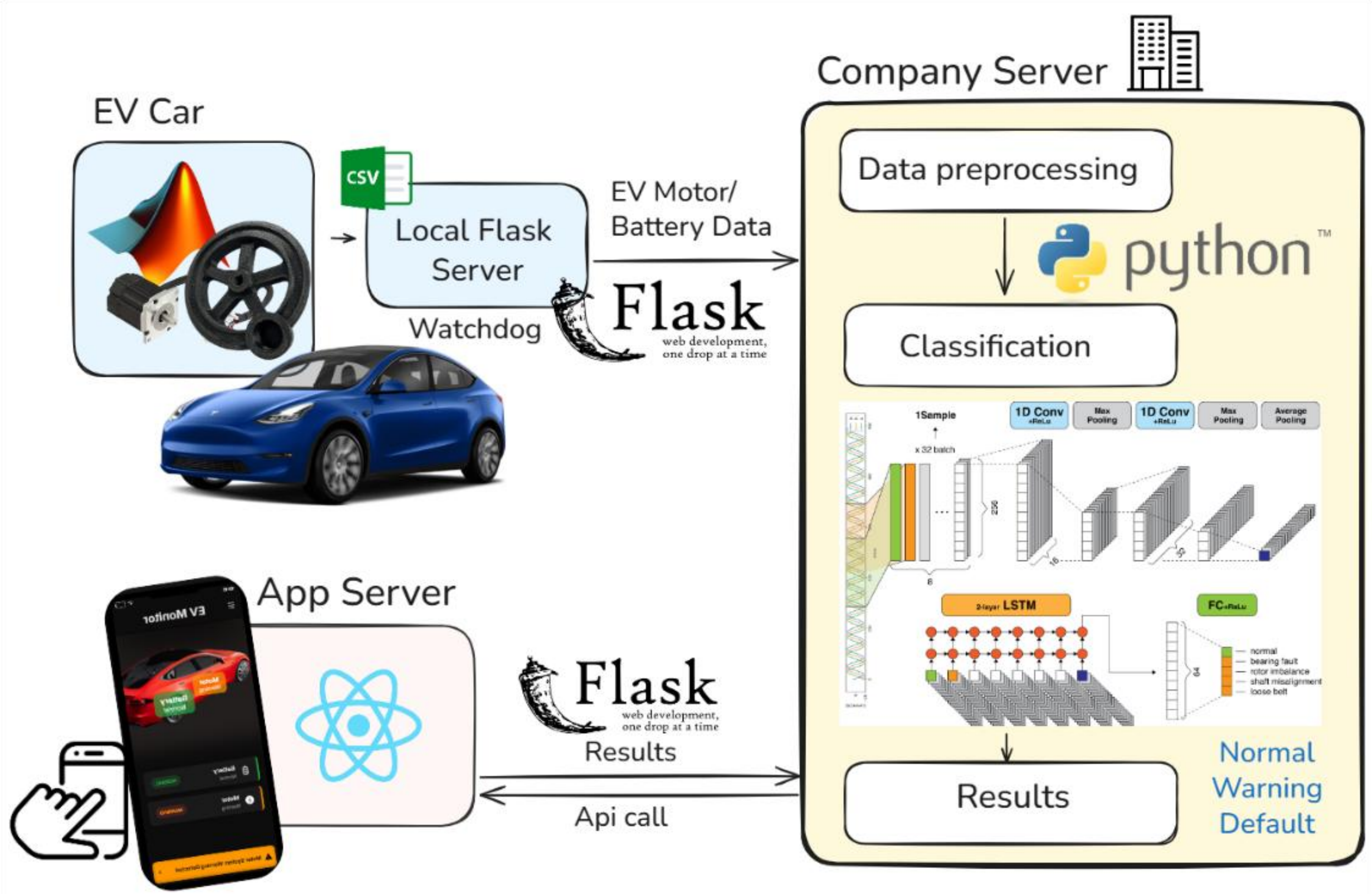
단독 **현대차 전기버스 '인휠 모터' 결함...판매분 전량 '무상 수리' 조치**

발행일 : 2020-05-28 11:23 지면 : 2020-05-29 3면

쇳가루·소음 발생 등 문제 지속 제기
일부 차량선 동력 구동 제동에도 영향
2018년 11월 이후 생산된 253대 대상
현대차 "개선품 적용…완성도 향상 최선"

Battery

Proposed Framework
for Smart Fault Detection

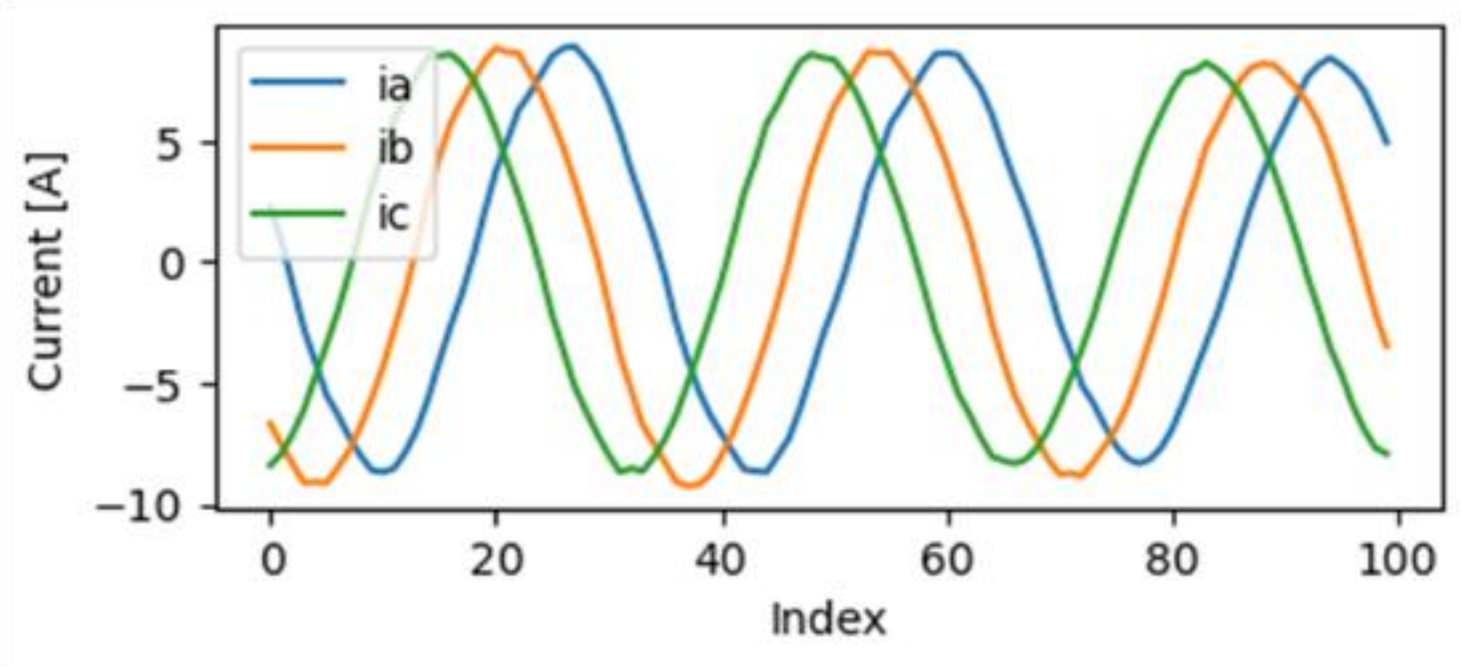


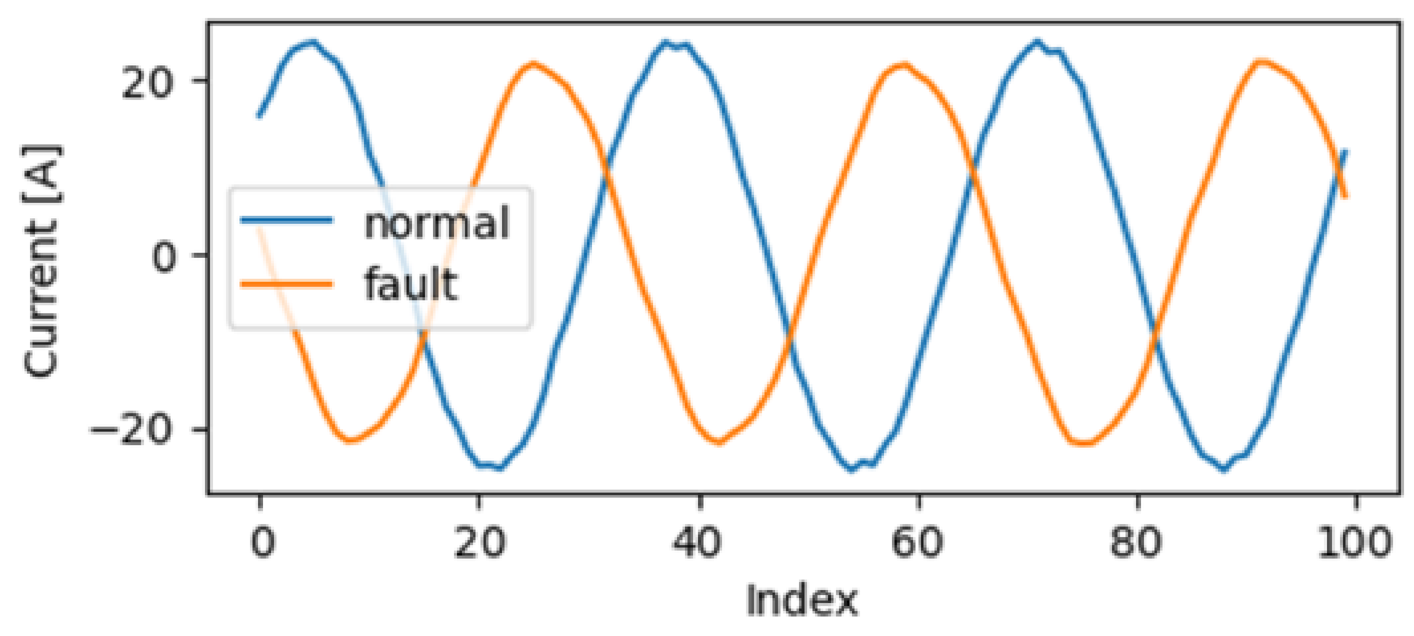
- **Daily Data Collection: Vehicle operation data (battery & motor signals) are recorded during driving.**
- **Cloud Diagnosis: Data is transmitted to the company server when the vehicle is idle. Deep learning models (LSTM, CNN–LSTM) classify potential faults.**
- **User Notification: Results are sent to the driver via a mobile application, enabling proactive maintenance.**
- **Goal: Provide an early-warning platform for EV fault detection, improving safety and reliability**

02; Method

Methodology

Dataset





Battery Model

- **Samples: 200k labeled time-series samples from 198 Evs**
- **Structure: Each sample → 128 time steps, 7 features (voltage, current, SOC, etc.)**
- **Distribution: 57.3% defective vs. 42.7% normal**

J. Zhang, Y. Wang, B. Jiang et al., "Realistic fault detection of li-ion battery via dynamical deep learning," Nature Communications, vol. 14, 2023

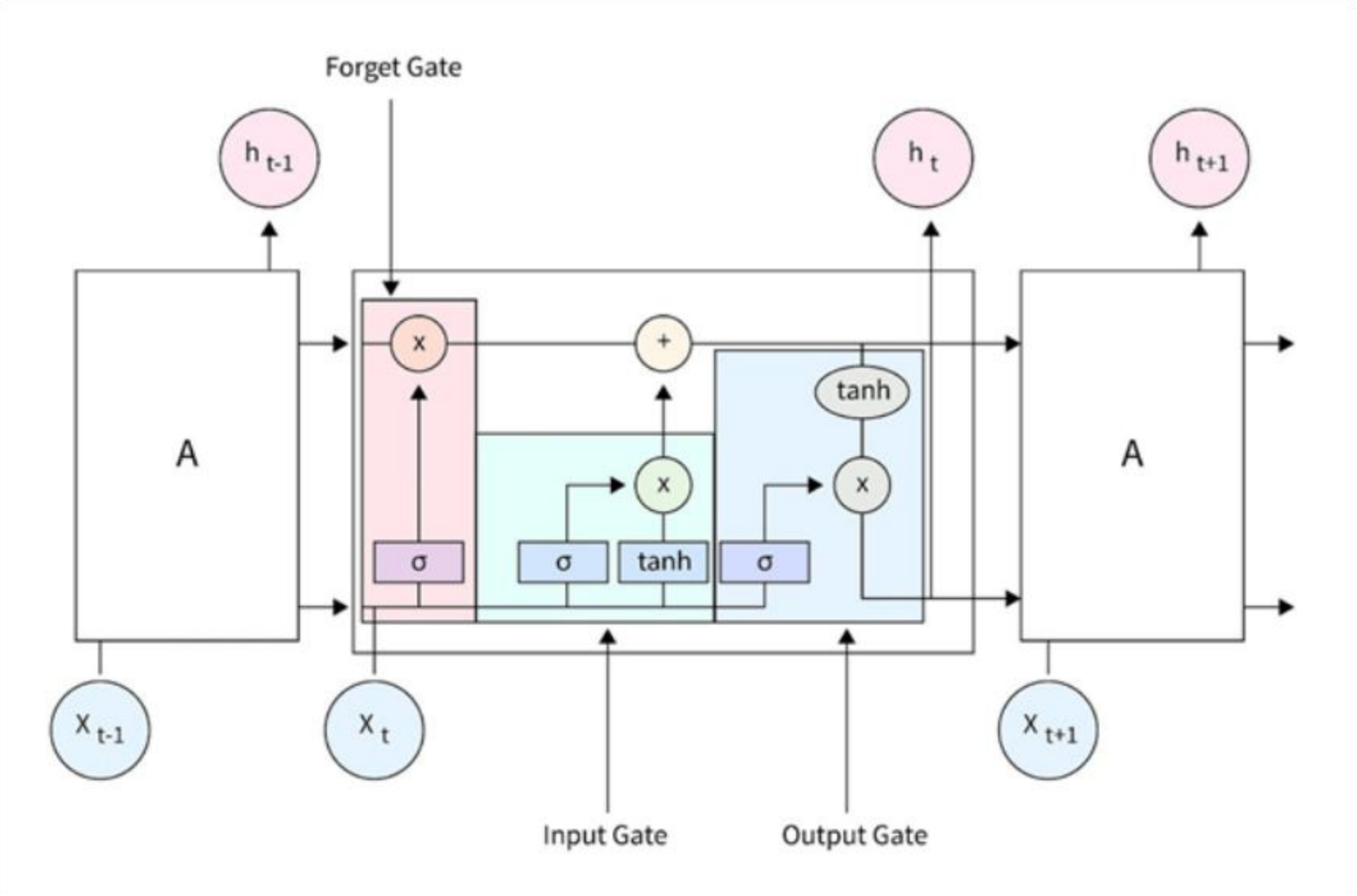
Motor Model

- **Hardware: Subway ventilation motors (5.5 kW and 11 kW)**
- **Signals: Three-phase current signals (Ia, Ib, Ic)**
- **Classes: 5 categories (Normal 50.9% , Rotor Unbalance 11.3%, Bearing Fault 15.1%, Loose Belt 11.3%, Sensor Fault 11.3%)**

2023. [2] AI Hub, "Machinery failure prediction sensor," https://aihub.or.kr, 2023, used for CNN-LSTM based motor fault detection

Methodology-Battery Model

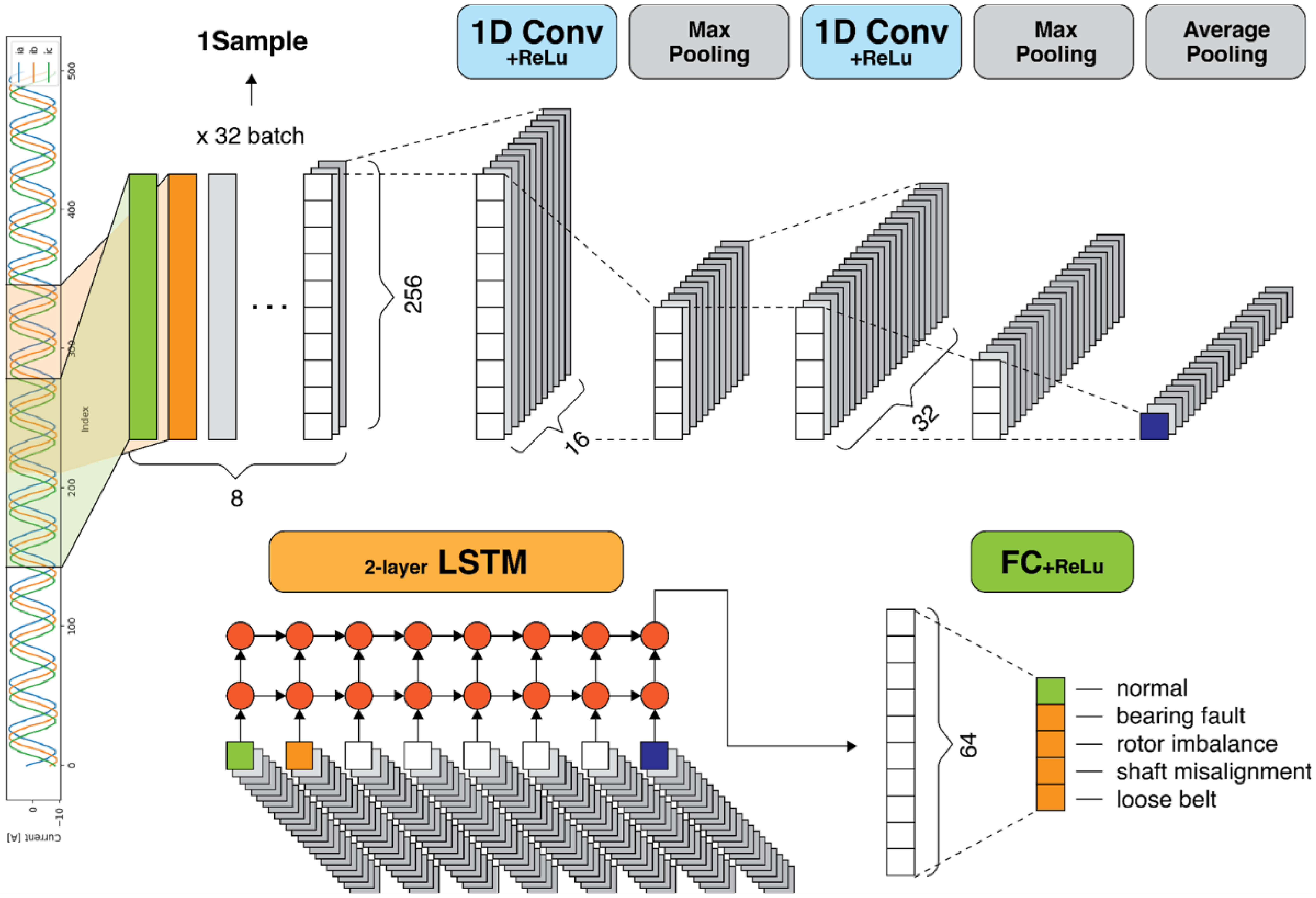
LSTM



Category	Hyperparameter	Value / Description
Preprocessing	Standard normalization	True
Training	Batch size	2048
	Epochs	25
Optimizer	Optimizer	Adam
	Learning rate	0.01
scheduler	LR scheduler	ReduceLROnPlateau (patience=3, factor= 0.5)
Loss	Loss function	BCEWithLogitLoss, pos_weight = normal_count/ default_co unt
Data Split	Train/Test split	8 : 2 (stratified)

Methodology-Motor Model

CNN + LSTM



Category	Hyperparameter	Value / Description
Preprocessing	Per-sample normalization	True
Training	Batch size	32
	Epochs	20
Optimizer	Optimizer	Adam
	Learning rate	0.001
	LR scheduler	ReduceLROnPlateau (patience=2, factor=0.5)
Loss	Loss function	CrossEntropyLoss
Normalization	BatchNorm usage	After each Conv1d layer
Data Split	Train/Val/Test split	Train/Val/Test 8:1:1 (stratified)

Motor set up

(a) (b) (c)

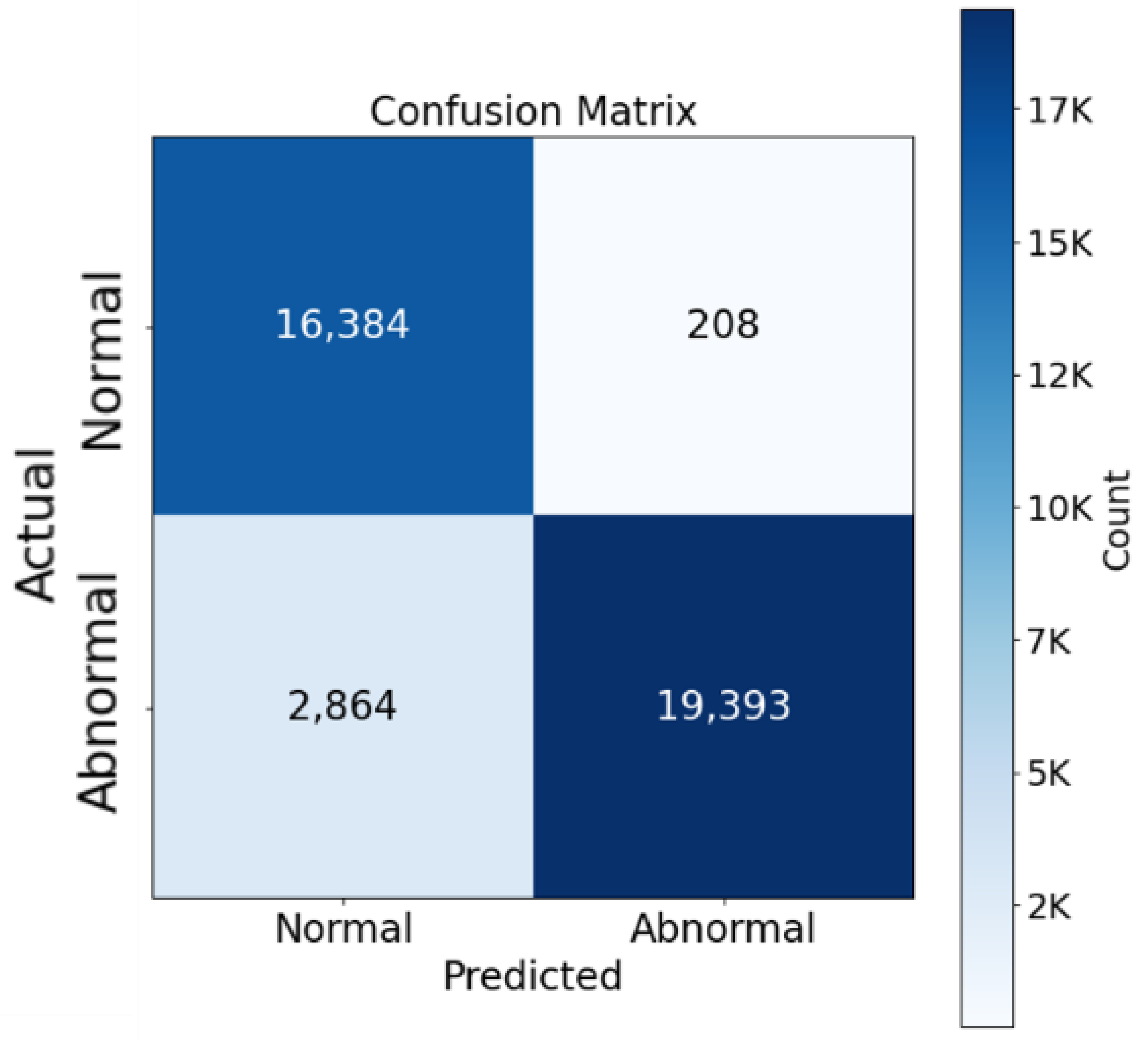
(a) Normal: 0 g (b) Warning: 50 g (rotor imbalance – mild)
(c) Fault: 100 g (rotor imbalance – severe)

- Permanent Magnet Synchronous Motor (PMSM) with custom 3D-printed wheel
- Controlled at constant 300 RPM using TI controller and inverter
- Fault simulation via attached weights
- Current signals (Ia, Ib, Ic) collected and stored for model inference

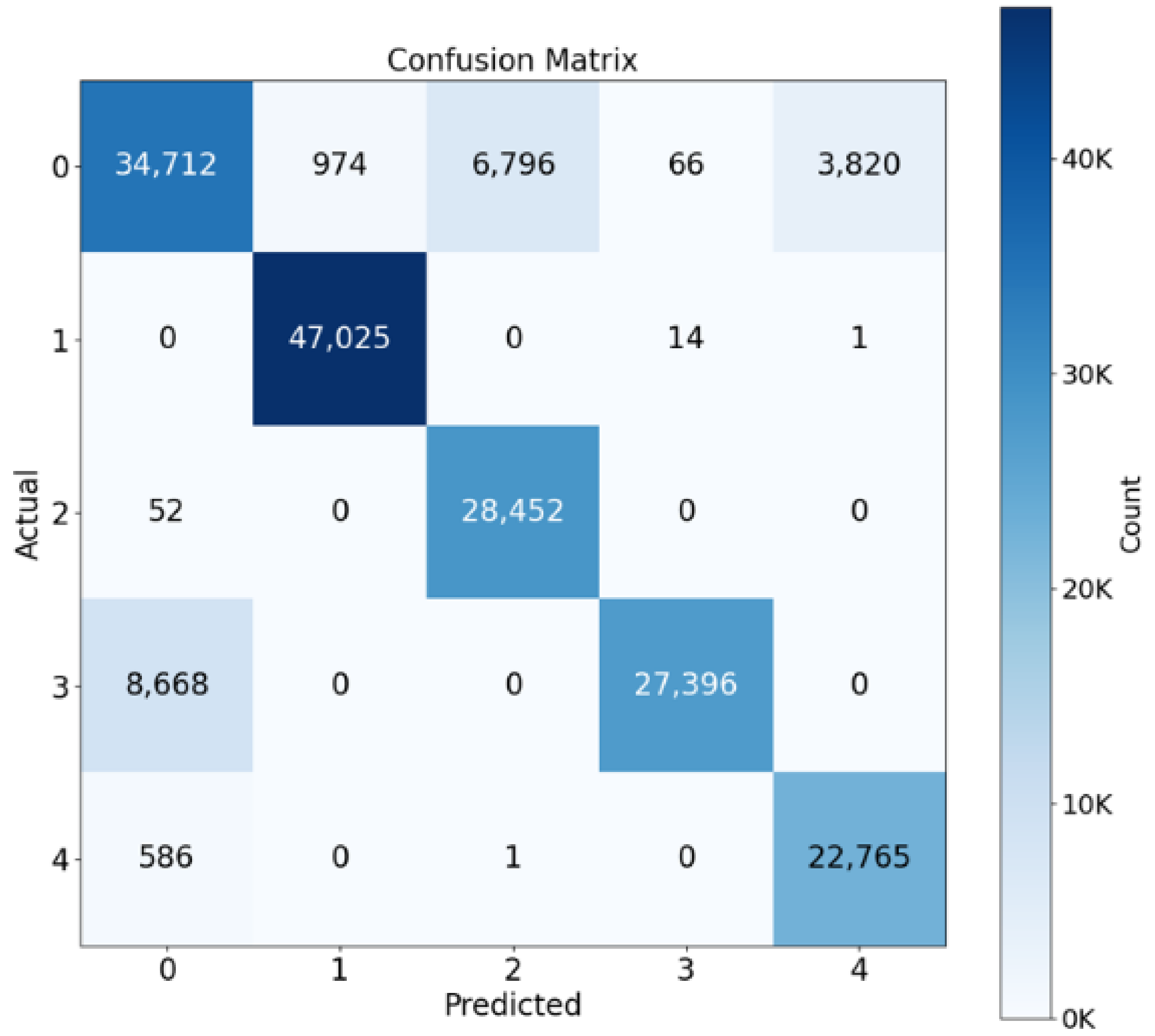
03; Results

Experiment Results

Results



(a)

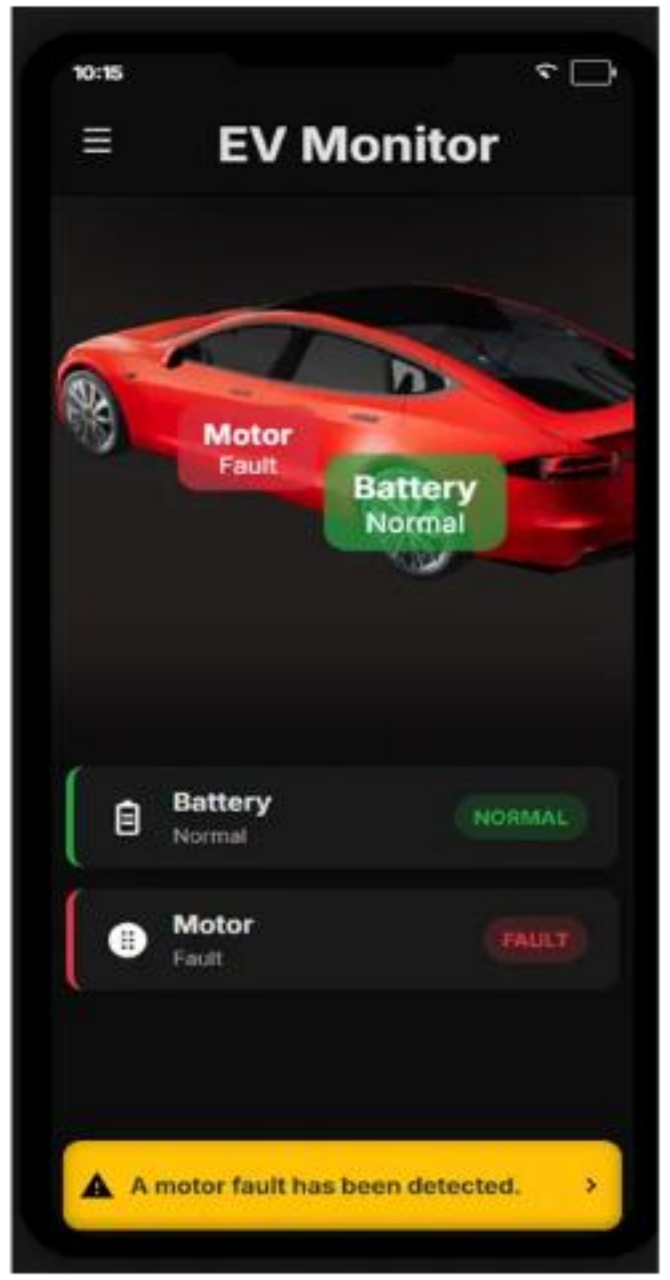


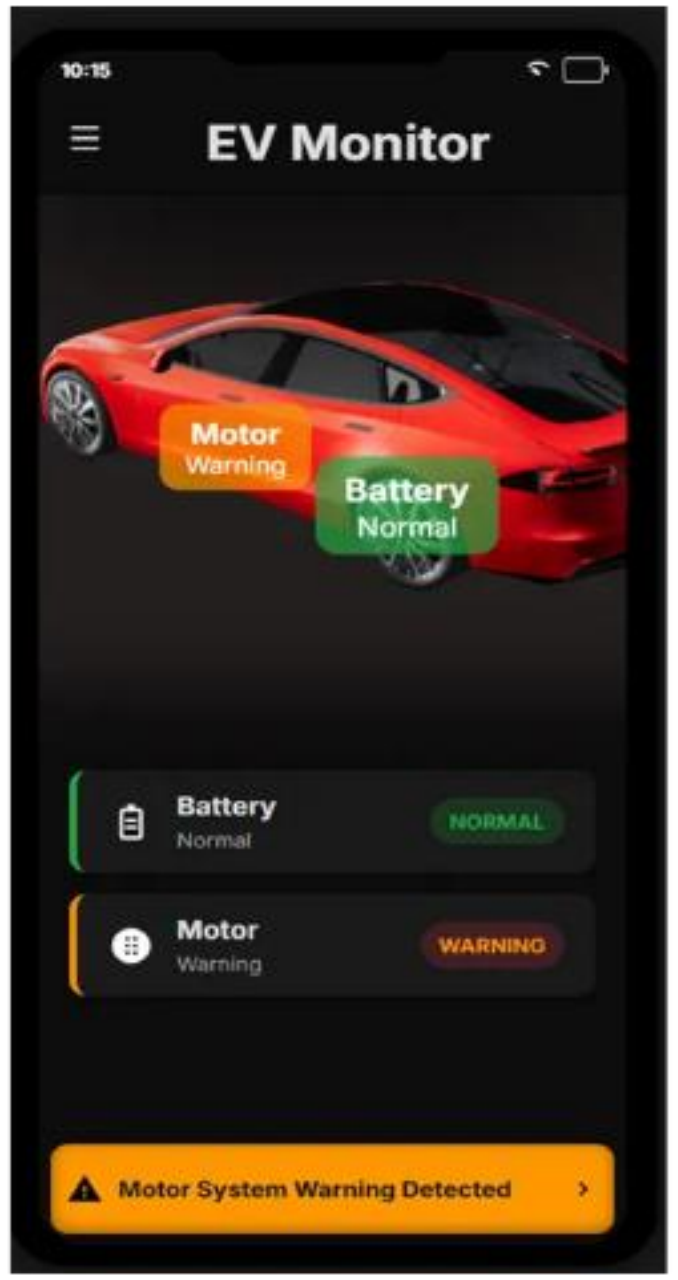
(b)

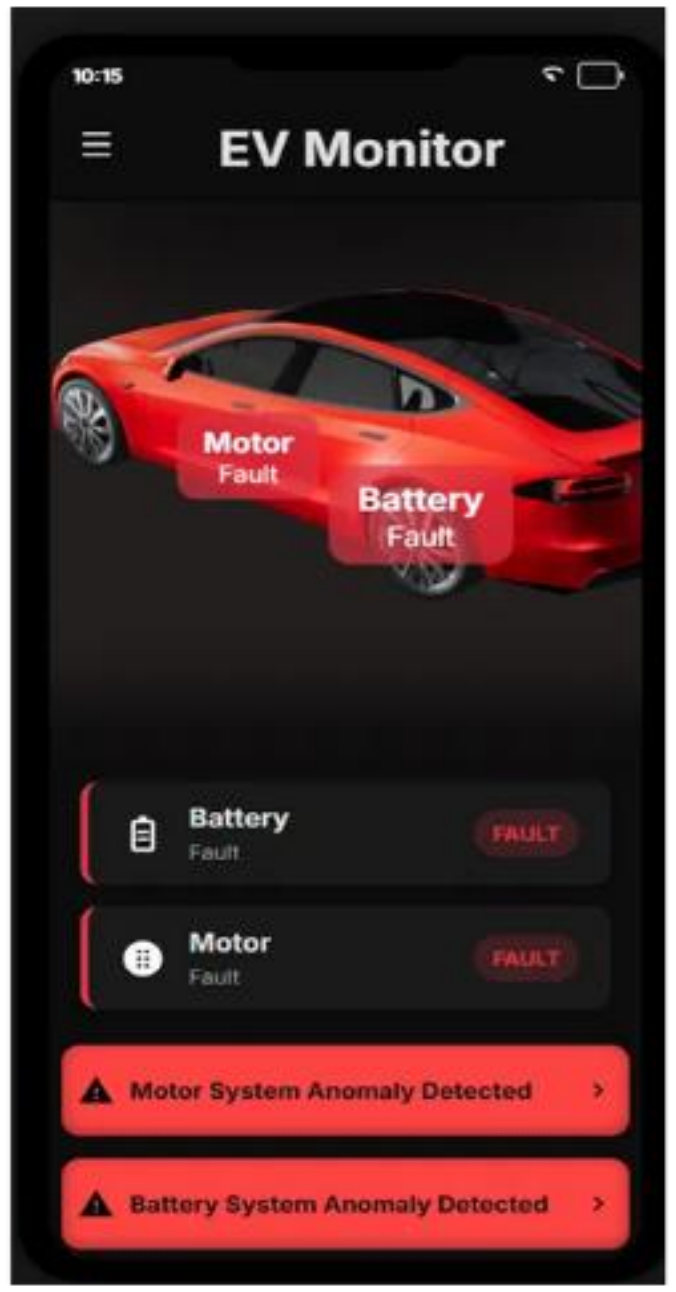
(a) Battery Fault Detection(Binary)
- Accuracy: ~92%
- F1-score: ~0.93

(b) Motor Fault Detection (Multi-class)
- Accuracy: ~90%
- AUC: >0.9 across all classes

Proof of Concept







04; Conclusion

Discussion

Discussion

- Contributions
  - Battery (binary) + Motor (multi-class) fault detection
  - CNN–LSTM model with >90% accuracy
  - Proof-of-concept with real motor setup & app
- Limitations
  - Battery: only public dataset used
  - Motor: simple imbalance case only
- Implication
  - Early fault detection can enhance EV safety and reliability

Thank you.
Q&A

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